

PROJECT TITLE : DIGITAL TOKEN MANAGEMENT
SYSTEM USING LPC2148

COURSE NAME: ARM PROCESSOR AND APPLICATION LAB
COURSE PROJECT

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Abstract :

The Digital Token Management System is designed to manage customer queues in places such as banks, hospitals, and service centers. The system uses the LPC2148 ARM7 microcontroller to generate and display token numbers on multiplexed seven-segment displays.

A timer-based delay mechanism is used for display multiplexing, while GPIO pins control the segment and digit selection lines. The token number is incremented automatically from 00 to 99 and displayed continuously. The project demonstrates embedded system concepts such as GPIO interfacing, multiplexing, display control, and real-time operation.

The implemented system successfully displays token numbers with minimal hardware and can be extended for keypad-controlled token generation

Introduction :

Queue management is an important requirement in public service environments. Manual token systems often lead to confusion and inefficient handling of customers.

A digital token management system provides an organized method for displaying token numbers. The system automatically updates and displays the current token number using electronic displays.

The LPC2148 microcontroller offers sufficient GPIO resources and processing capability to implement such systems efficiently.

Objective:

- Design a digital token display system using LPC2148.
- Interface multiplexed seven-segment displays.
- Generate sequential token numbers.
- Demonstrate embedded display control techniques.
- Provide a foundation for future queue management applications.

Problem Statement :

In crowded service environments, manual queue handling creates confusion and delays.

The objective of this project is to design and implement a Digital Token Management System that generates and displays token numbers from 00 to 99 using multiplexed seven-segment displays controlled by the LPC2148 microcontroller.

System Overview :

The system operates on a simple sequential structure:

Input

- Internal counter value
- Future enhancement: Push button or keypad

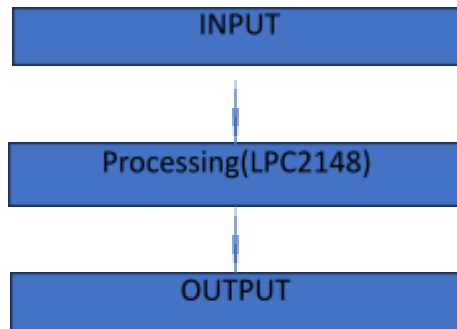
Processing

- LPC2148 increments token number.

- Number is separated into tens and units digits.
- Multiplexing logic alternately drives both displays.

Output

- Two multiplexed seven-segment displays showing token numbers.



Working Principle :

The microcontroller continuously increments the token count from 00 to 99. Each number is divided into tens and units digits. The corresponding segment patterns are sent to the seven-segment displays. Through rapid multiplexing, both digits appear continuously illuminated.

6. PROBLEM ANALYSIS

Features

- Displays token numbers from 00–99.
- Uses multiplexing to reduce GPIO usage.
- Simple and cost-effective design.
- Real-time display operation.

Assumptions

- Supply voltage = 3.3V.
- Seven-segment displays operate correctly with GPIO outputs.
- Token count range limited to 00–99.
- Stable power supply available.

Justification

Why LPC2148?

- RISC architecture.
- Multiple GPIO pins.
- Suitable for display interfacing.
- Widely used in embedded laboratories.

Why Multiplexed Display?

- Reduces hardware requirements.
- Saves microcontroller pins.
- Provides efficient display control.

Peripherals Used

On-Chip Peripherals

1. GPIO
2. Timer (Delay generation)

Off-Chip Peripherals

1. Seven Segment Display 1
2. Seven Segment Display 2

Hardware Description :

LPC2148 Microcontroller

Features:

- ARM7 RISC Architecture
- 32-bit architecture
- 60 MHz clock
- Multiple GPIO ports
- Timers and communication peripherals

Seven Segment Displays

A seven-segment display consists of seven LEDs arranged in the shape of the number "8".

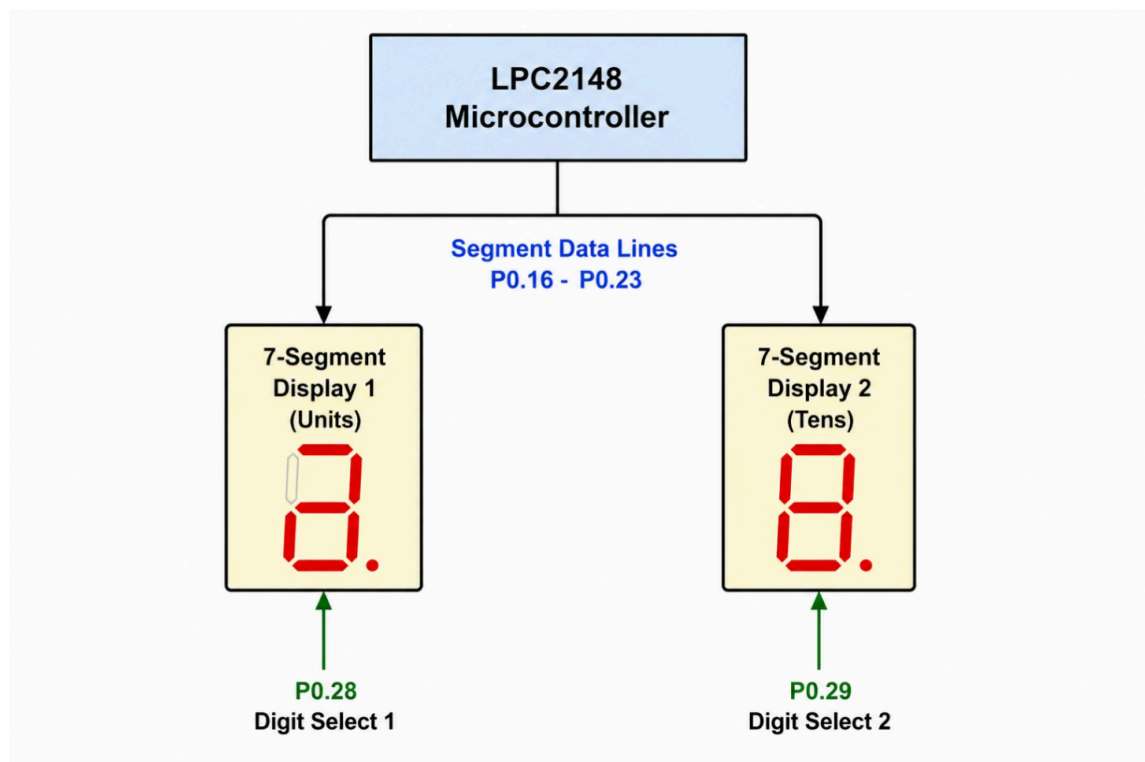
Segments

a, b, c, d, e, f, g, h

Display Type

Common Cathode (according to segment code used)

Block Diagram :



Components Required

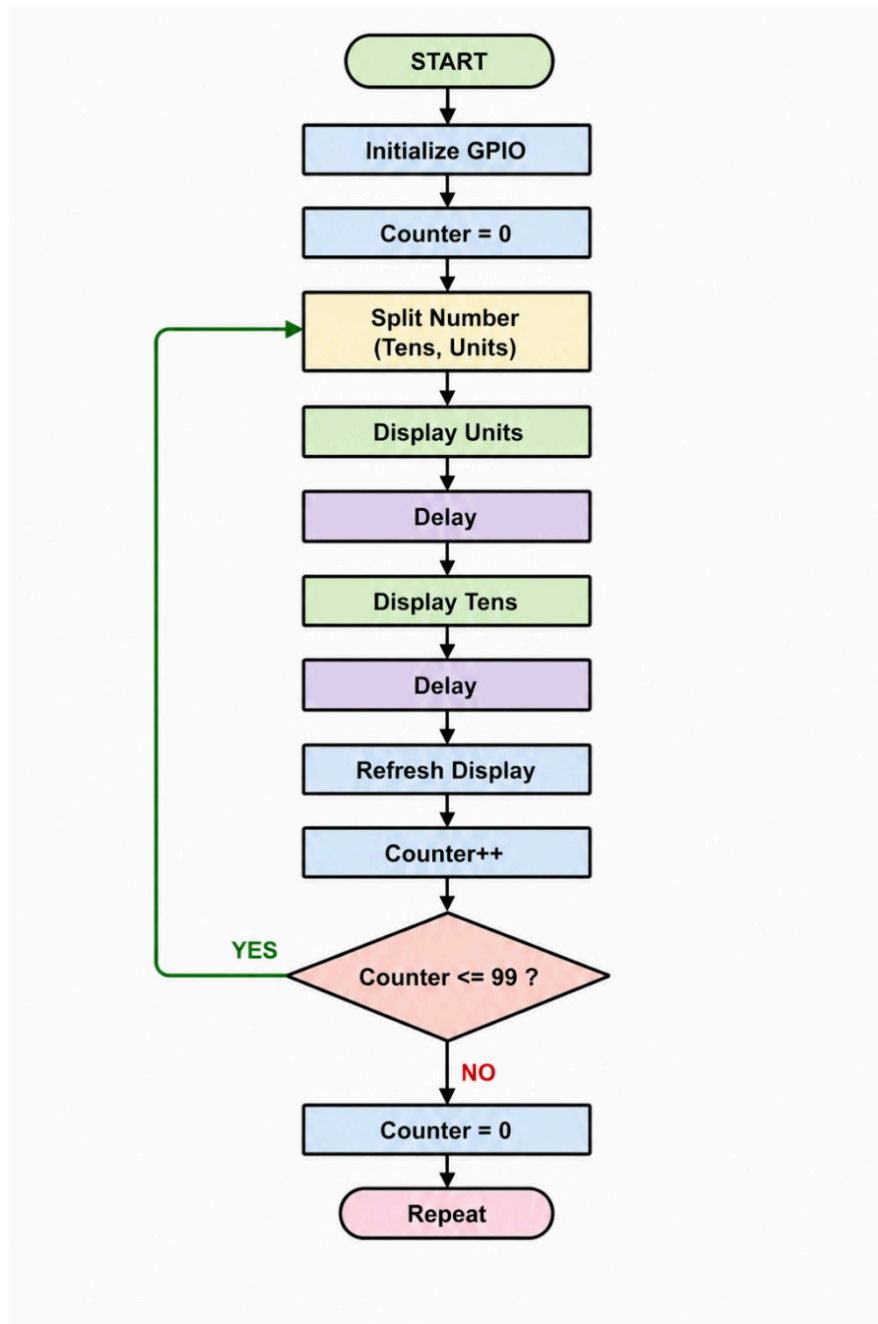
Component	Quantity
LPC2148 Board	1
Seven Segment Display	2
Resistors	14
Connecting Wires	As required

Component

Quantity

Algorithm / Flowchart :

Flowchart :



Algorithm :

Step 1: Initialize PINSEL0 and PINSEL1 to configure all pins as GPIO.

Step 2: Configure P0.16–P0.23 and P0.28–P0.29 as output pins.

Step 3: Start the counter from 0.

Step 4: Separate the counter value into tens and units digits.

Step 5: Clear segment data and digit select lines.

Step 6: Display units digit by enabling Display 1.

Step 7: Introduce a short delay.

Step 8: Clear display lines.

Step 9: Display tens digit by enabling Display 2.

Step 10: Introduce a short delay.

Step 11: Repeat multiplexing continuously.

Step 12: Increment counter until 99.

Step 13: Restart from 00.

CODE:

```
#include<lpc21xx.h>
void delay(unsigned int);
void disp(unsigned int);
int main()
{
    unsigned long int value,i;
    unsigned int row0[4]={ 0x00ee0000,0x00ed0000,0x00eb0000,0x00e70000};
    unsigned int row1[4]={ 0x00de0000,0x00dd0000,0x00db0000,0x00d70000};
    unsigned int row2[4]={ 0x00be0000,0x00bd0000,0x00bb0000,0x00b70000};
    unsigned int row3[4]={ 0x007e0000,0x007d0000,0x007b0000,0x00770000};
    IO1DIR  = 0xFFFF0FFF;
    PINSEL1=0x00000000;
    IODIR0=0xf0ff0000;
    IOSET0=0XF0000000;
    while(1)
    {
        IO1PIN=0x00ff0000;
        IOCLR1=0x00100000;
        value=IOPIN1;
        delay(50000);
        value=value & 0x00ff0000;
        for(i=0; i<4;i++)
        {
```



```

        if(value==row0[i])
        {
            disp(i);
            delay(65000);
                delay(65000);
            //delay(65000);delay(65000);delay(65000);
        }
    }
IO1PIN=0x00ff0000;
IOCLR1=0x00200000;
value=IOPIN1;
delay(50000);delay(50000);
value=value & 0x00ff0000;
for(i=0; i<4;i++)
{
    if(value==row1[i])
    {
        disp(i+4);
        delay(65000);delay(65000);
        //delay(65000);delay(65000);
    }
}
IO1PIN=0x00ff0000;
IOCLR1=0x00400000;
value=IOPIN1;
delay(65000);delay(65000);delay(65000);
delay(65000);delay(65000);
value=value & 0x00ff0000;
for(i=0; i<4;i++)
{
    if(value==row2[i])
    {
        disp(i+8);
        delay(50000);
    }
}

```

```

    }
}
IO1PIN=0x00ff0000;
IOCLR1=0x00800000;
value=IOPIN1;
delay(65000); delay(65000);
value=value & 0x00ff0000;
for(i=0;i<4;i++)
{
    if(value==row3[i])
    {
        disp(i+12);
        delay(65000);delay(65000);
        //delay(65000);delay(65000);
    }
}
}
}

```

```

void disp(unsigned int temp)
{
    unsigned int i;
    unsigned int da[16]={0xf03F0000, 0xf0060000, 0x305B0000, 0x304F0000,
    0x00660000,0x006D0000,
                                0x007D0000, 0x00070000, 0x007F0000,
    0x006F0000, 0x00770000,0x007C0000,
                                0x00390000, 0x005E0000, 0x00790000,
    0x00710000 };
}

```

```

IOCLR0=0x00ff0000;
IOSET0=0x10000000;
i=temp;
IOSET0|=da[i];
delay(65000);delay(65000);delay(65000);
delay(65000);delay(65000);

```

```
//IOCLR0=0X00FF0000;
}
```

```
void delay(unsigned int del)
{ unsigned int k;
  for(k=0;k<del;k++);
}
```

10. INTERFACING DETAILS

10. INTERFACING DETAILS	
LPC2148 Pin	Connection
P0.16	Segment a 
P0.17	Segment b 
P0.18	Segment c 
P0.19	Segment d 
P0.20	Segment e 
P0.21	Segment f 
P0.22	Segment g 
P0.23	Decimal Point 
P0.28	Display 1 Select  Display 1 (Units)
P0.29	Display 2 Select  Display 2 (Tens)

11. CALCULATIONS & SPECIFICATIONS

Counter Range

Minimum Token = 00

Maximum Token = 99

Total Tokens:

99 - 00 + 1
= 100 Tokens

Display Refresh

Each digit delay:

delay(5000)

Two displays:

2 × delay(5000)

Rapid switching creates persistence of vision.

GPIO Usage

Segment Lines:

8 GPIO Pins

Digit Select Lines:

2 GPIO Pins

Total:

10 GPIO Pins

12. IMPLEMENTATION

Software

- Keil µVision IDE
- Embedded C

Hardware

- LPC2148 Development Board
- Dual Seven Segment Display

Program Description

The program uses an array containing segment codes for digits 0–9. The display() function separates the token number into tens and units digits and performs multiplexing. The main loop increments token numbers from 00 to 99 and displays them continuously.

13. RESULTS

Expected Output

00
01
02
03
...
98
99

Observations

- Stable display obtained.
- Proper multiplexing achieved.
- Token count increments correctly.
- No visible flickering.

Conclusion :

The Digital Token Management System using LPC2148 was successfully implemented. The system displays token numbers from 00 to 99 using multiplexed seven-segment displays. The project demonstrates GPIO interfacing, display multiplexing, and embedded control techniques. The obtained results verify correct operation of the designed system.

Future Scope :

Keypad-based token generation.

- Push button token increment.
- LCD display integration.
- Wireless token display system.
- Voice announcement of token numbers.
- UART communication with central server.
- Multiple counter management.

16. REFERENCES

1. LPC2148 User Manual.

2. ARM7TDMI-S Technical Reference Manual.
3. Embedded Systems by Raj Kamal.
4. Keil μ Vision Documentation.
5. NXP LPC2148 Datasheet.
6. Seven Segment Display Datasheet.